

Article Review – Wearable Healthcare Devices Enabled by 5G Communication

Question 21: Wearable Healthcare Devices Enabled by 5G Communication

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Base Article: 5G Technology in Healthcare and Wearable Devices: A Review

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Article Review

The article examines how 5G communication can support healthcare and wearable devices. The main focus is on applications such as remote patient monitoring, chronic disease management, the Internet of Medical Things (IoMT), telemedicine and robotic surgery. The authors review earlier studies to explain where the features of 5G can be useful in healthcare.

The paper explains that high data capacity, low latency and the ability to connect many devices make 5G suitable for continuous health monitoring. Wearable sensors can collect patient information and transmit it to healthcare systems, allowing doctors or caregivers to monitor patients remotely. The review also discusses the use of local or edge processing, which can help health information be handled closer to the device when a quick response is required.

One useful part of the paper is that it does not present 5G as a complete solution by itself. The authors also discuss practical problems with wearable healthcare systems. These include limited battery capacity, possible sensor inaccuracies, security and privacy concerns, infrastructure cost and differences in network availability. These issues are important because healthcare devices may need to operate continuously and handle sensitive patient information.

In my view, the main strength of the article is its broad view of the connection between 5G, wearable devices and healthcare applications. Since the paper is a review, it brings together previous work rather than testing one new wearable system. Therefore, it gives a good understanding of the field, but it does not provide one experimental system that can be directly evaluated. Overall, the paper shows that 5G can improve the communication side of wearable healthcare, but reliable sensors, long battery life, security, privacy and affordable infrastructure are also necessary for practical use.

Reference

5G Technology in Healthcare and Wearable Devices: A Review. Delshi Howsalya Devi, Kumutha Duraisamy, Ammar Armghan, Meshari Alsharari, Khaled Aliqab, Vishal Sorathiya, Sudipta Das and Nasr Rashid (2023). Sensors, 23(5), 2519. DOI: 10.3390/s23052519.